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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG
Department of Computer Science & Engineering (CSE)
Mid-Term Examination, Autumn-2015

Course Code: CSE-4801 Course Title: Compiler

Full Marks: 30 Time: 1 Hour 30 minutes

[Answer any three of the following questions. The figure in the margin illustrate marks] *3*10=30*

1. a) What is compiler? Write down the differences between compiler and interpreter. 2

b) What do you mean by analysis of a source program? Write down the three analysis phases in compiling a source program. 2

c) Explain the translation of following statement with figure by using compiler phases: 4
`z= x*y-5`

d) What do mean by synthesized attribute? Why is it required? 2

2. a) Consider following grammar: 1+1

$E \rightarrow E + T$
$E \rightarrow E - T$
$E \rightarrow T$
$T \rightarrow 0|1|2|3|4|5|6|7|8|9$

Draw the abstract syntax tree (AST) and parse tree for the input "10-2+5-1"

b) Consider following grammar: 1+1

$E \rightarrow +EE$
$E \rightarrow *EE$
$E \rightarrow -EE$
$E \rightarrow x|y$

Find the leftmost derivation and rightmost derivation for the string "+*-xyxy".

c) Define the right-linear and context sensitive grammar. Classify the following grammar as regular, left-linear, right-linear, context free or context sensitive. Some grammar may fit more than one category. 1+2

G1:
S → aAb
aA → aB
aB → aaA
B → a

G2:
$S \rightarrow aA|\varepsilon$
$A \rightarrow aA$
$A \rightarrow bB$
$B \rightarrow bB|b$

G3:
$S \rightarrow aSa|\varepsilon$
$B \rightarrow bSb$
$B \rightarrow b$
$C \rightarrow c$

G4:
$S \rightarrow aS$
$S \rightarrow aSbS$
$S \rightarrow \varepsilon$

d) Define ambiguity. Determine following grammars are ambiguous or not. Justify your answer. 3

$Expr \rightarrow Expr + Expr$
$Expr \rightarrow Term$
$Term \rightarrow Term * Term$
$Term \rightarrow (Expr)$
$Term \rightarrow a$

$Stmt \rightarrow if\ Expr\ then\ Stmt$
$Stmt \rightarrow if\ Expr\ then\ Stmt\ else\ Stmt$
$Stmt \rightarrow a$
$Expr \rightarrow b$

3. a) Explain the role of lexical analyzer with block diagram. Give the tokens, lexemes and patterns of the C programming statement `int result=a+b-60;`. 3

b) Give a regular expression to define the identifier of C programming language. 2

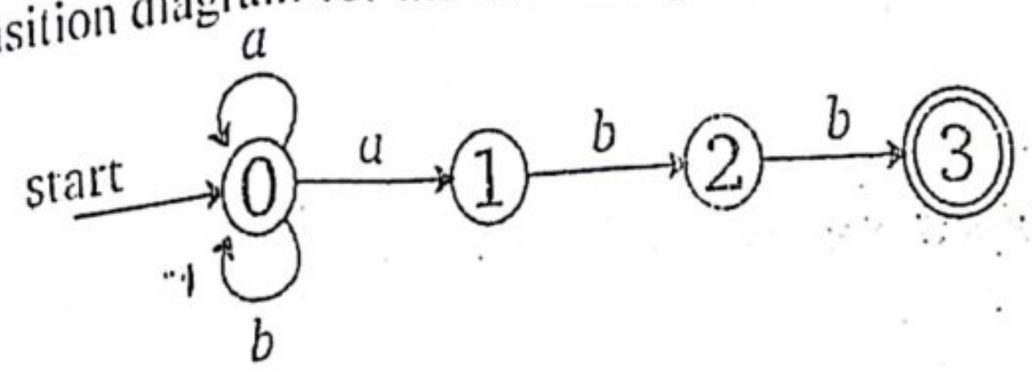
c) Construct the NFA or $\varepsilon - NFA$ for the following regular expression: 2.5

$(a|b)^*a(a|b)$

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d) Draw the DFA transition diagram for the following NFA:



4. a) Show the phases of compiler with figure. Mention the some names of useful compiler construction tools.

b) For the following gramntársG1 identify useless non-terminals and find the unreachable non-terminals from G2. Give the modified grammars G1 with no useless non-terminals and G2 without unreachable non-terminals but with same language as original

G1:

$S \rightarrow iEtS$

$S \rightarrow iEtSeS$

$S \rightarrow a$

$E \rightarrow (S)$

G2:

$S \rightarrow aA$

$A \rightarrow a|b$

$B \rightarrow bC$

$C \rightarrow a$

c) Consider following CFG G=({X},{0,1},P,X):

$X \rightarrow \varepsilon$

$X \rightarrow 0|1$

$X \rightarrow 0X0$

$X \rightarrow 1X1$

What language L(G) is generated by this G? Justify your answer.

d) Define the left recursion and context free grammar with example.

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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science &Engineering (CSE)
Mid-Term Examination, Autumn, 2017
Course Code: CSE-4801 Course Title: Compiler
Full Marks: 30 Time: 1 Hour 30 minutes

[Answer any three questions from the following four questions.]

1. a) Define compiler .What is the role of compiler? 2
   b) Illustrate briefly different phase of compiler by considering following example. 4
   Clearly indicate the output of each phase with figure.
   `current := initial + rate * 60;`
   c) What is the difference between compiler and interpreter .Describe the compiler construction tools. 4

2.a) Define useless nonterminal. Show the structure of compiler front end. 3

b) Explain context free grammar with example. 2

c) Consider the CFG: 3
$S \rightarrow +SS \mid *SS \mid S \mid a$
1) Show that: *aa+a** can be generated by the grammar.
2) Draw the parse tree.

d) Consider the grammar: 2
$P \rightarrow aPbP \mid bPaP \mid \varepsilon.$
Show that this grammar is ambiguous by constructing two different left most derivation for the expression *abab*.

3.a) Write down the differences between DFA and NFA. 2

b) Briefly describe the issue of lexical analysis. 2

c) Give the regular expression to define the identifier and an unsigned number (1.89E-4) of C programming language. 3

d) Construct DFA ,NFA and $\epsilon - NFA$ for the following regular expression: 3
$(a|b) * abb$

4.a) Define Token, Patterns and lexemes with example. 3

b) Elaborate the role of lexical analyzer. 3

c) Differentiate between context free grammar and context sensitive grammar. 2

d) Write down the following language into regular expression-- 2
1) Set of strings consisting of even number of a's followed by odd number of b's .
2) Set of strings consisting of zero or more *a*'s followed by a *b*.

END

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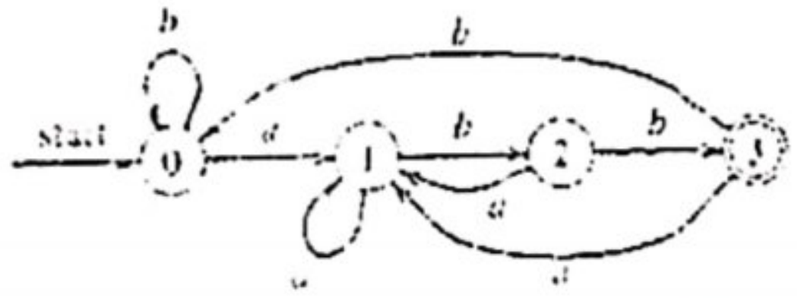
INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG
Department of Compater Science &Engineering (CSE)
Mid-Term Examination, Spring, 2017
Course Code: CSE-4801 Course Title: Compiler
Full Marks: 30 Time: 1 Hour 30 minutes

[Answer any three questions from the following four questions.]

1. a) Define compilers .What is the role of compiler? 2
   b) Illustrate briefly different phase of compiler by considering following example. 5
   (Clearly indicate the output of each phase with figure.
   `position := current-initial + rate * 0.05;`
   c) Show the components of a language processing system with necessary figure. 3
   Write down the analysis phase of the source program at compiling.

2.a) Explain context free grammar with example. 2
   b) Define lexeme and pattern Give the tokens, lexemes and patterns of the following 3
   C programming statement:
   `float position := current-initial + rate * 0.05;`
   c) Consider the CFG. 3
   $E \rightarrow E+E|E*E|E--E|-E|E|id$
   i). Show that: *-id +id*id* can be generated by the grammar.
   ii. Draw abstract syntax tree and the parse tree. 2
   d) Consider the grammar:
   $P \rightarrow aPbP|bPaP|\varepsilon$..
   Show that this grammar is ambiguous by constructing two different left most derivation for the expression *abab*.

3.a) Write down the differences between DFA and NFA. 2
   b) Give the regular expression and language of the following DFA: 2



   c) Give the regular expression and construct the *NFA* and *DFA* to define the identifier of C programming language. 3
   d) Construct NFA and $\varepsilon - NFA$ for the following regular expression: 3
   (a|b) $a(a|b)(a|b)$

4.a) Elaborate the functions of lexical analyzer 3
   b) Show the frontend and backend phases of compiler with figure. 3
   c) What do you mean by the right-linear and context sensitive grammar?. 2
   d) Write down the following language into regular expression-- 2
   1) set of strings consisting of even number of a's followed by odd number of b's .
   2)set of strings of 1's and 0's of even length..

END

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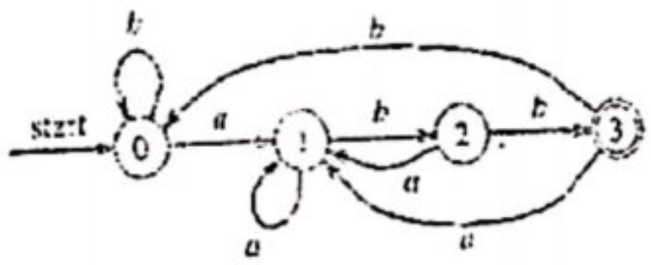
INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG
Department of Computer Science & Engineering (CSE)
Mid-Term Examination, Autumn, 2018
Course Code: CSE-4801 Course Title: Compiler
Full Marks: 30 Time: 1 Hour 30 minutes

[Answer any three questions from the following four questions.]

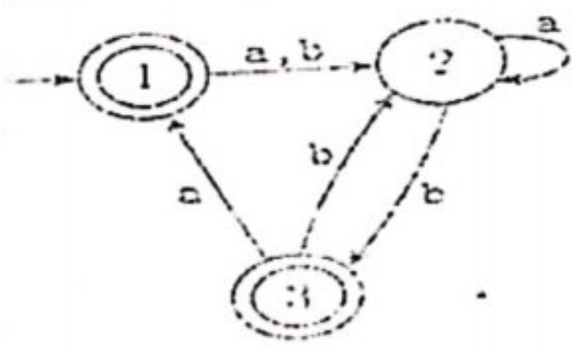
1. a) Define compilers and interpreter .What is the role of compiler? 2
   b) Illustrate briefly different phase of compiler by considering following example. Clearly indicate the output of each phase with figure. 5
   Grade := total_grade_points * 163 / total_credit;
   c) Compose the advantages of language processing system in which the compiler produces assembly language rather than machine language? 3

2.a) Explain context free grammar and Context sensitive grammar with example. 3
   b) Define lexeme and pattern. Give the tokens, lexemes and patterns of the following C programming statement: 3
   a:= b*c-d
   c) Consider the following CFG: 2
   $E \rightarrow +EE \quad E \rightarrow -EE$
   $E \rightarrow *EE \quad E \rightarrow xly$
   Find the left most derivation and right most derivation for the string
   +-*xyxy
   d) Consider the grammar: 2
   $P \rightarrow aPbP|bPaP|\varepsilon.$
   Show that this grammar is ambiguous by constructing two different left most derivation for the expression *abab*.

3.a) Express the main idea of NFA and discuss with exampies(a/b) 2
   b) Give the regular expression and language of the following DFA: 2



   c) Convert following regular expression into DFA (a/b)*baa(a/b)* 3
   d) Convert the following NFA to DFA 3



4.a) Elaborate the functions of lexical analyzer. 3
   b) Construct NFA and $\varepsilon - NFA$ for the following regular expression: 3
   $:(a|b)*a(a|b)(a|b)$
   c) What do you mean by the left-linear and left-linear grammar? 2
   d) Write down the following language into regular expression-- 2
   1) set of strings consisting of even number of a's followed by odd number of b's .
   2) set of strings of 1's and 0's of even length.

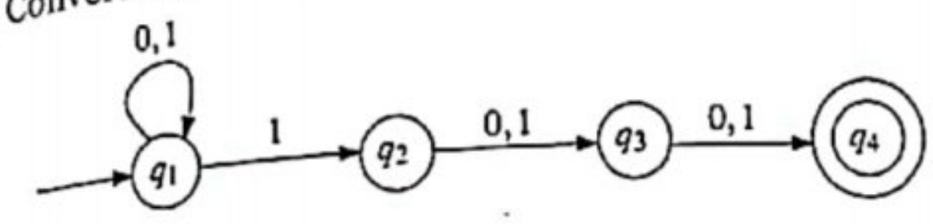
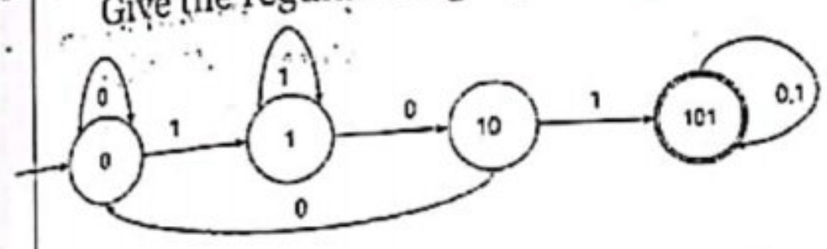
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International Islamic University Chittagong

Department of Computer Science & Engineering
B.Sc. in CSE, Mid Term Examination, Autumn 2019
Course Code: CSE-3527 **Course Title:** Compiler
Total Marks: 30 **Time:** 90 minutes

Answer any 3 from the following Questions. Figures in the right-hand margin indicate full marks

#	Questions	M	COs	DI
1	(a) Define *token, lexeme,* and *pattern.* Give the tokens, lexemes, and pattern for the following statement: **r = x * y – z;**	1 2	CO2	C1 C3
	(b) Convert following NFA to DFA and test generated DFA using proper string: 	4	CO2	C4
	(c) Describe the role of *front end* and *back end* of a compiler exploring all stages of them.	3	CO1	C2
2	(a) Write the differences between compiler and interpreter. Give the regular languages of the following DFA: 	2 2	CO2	C1 C3
	(b) Write down the following languages into the regular expression: 1. Set of strings consisting of even number of a's followed by odd number of b's 2. Set of strings of a's and b's of any length 3. Set of strings of 0's and 1's ending with the string **011**	3	CO2	C2
	(c) What is a language processing system? How many components are present in the language processing system, explain briefly with a required figure?	1 2	CO1	C1
3	(a) Consider the following context-free grammar: **X→ X X + | XX * | a** Show how the string **aaa+a*+aa**** can be generated by this grammar. Construct a parse tree for this string.	4	CO2	C4
	(b) Demonstrate briefly different phases of the compiler by considering the following example. Point out outputs of each phase with a figure.	3	CO1	C2
	(c) What is *left linear grammar* and *right linear grammar?* Describe whether the following grammar is ambiguous or not by considering the string 9 – 5 +1 **X → X + X | X – X | 0|1|2|3|4|5|6|7|8|9**	3	CO2	C1 C4
4	(a) Construct a DFA and a €-NFA for the following regular expression with step by step clarifications: ***(a|b)*a (a | b)***	4	CO2	C3
	(b) Find out the useless non-terminals, left linear grammar, Right linear grammar and context sensitive grammar from the following grammar:	3	CO2	C2

G1:αB → ab B →0B	G2: A →A0|A1|B00 B →B0|B1|€	G3:S →10S| 11S| C| € C →C1

	(c) (b) What is *bytecode*? Explain how Java language processors combine compilation and interpretation using figures.	3	CO1 C2

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50.

International Islamic University Chittagong

Department of Computer Science & Engineering
B.Sc. in CSE, Mid Term Examination. Autumn 2019
Course Code: CSE-4301 Course Title: Compiler
Total Marks: 30 Time: 90 minutes

Answer any 3 from the following Questions. Figures in the right-hand margin indicate full marks

#	Questions	M	COs	DL
1	(a) What is language processing system? How many components are exist in language processing system? Explain briefly with necessary figure.	1+2	CO1	C1
	(b) What is context sensitive grammar? Find out the useless non-terminals, left linear grammar, Right linear grammar and context sensitive grammar from the following grammar: **G1:αAβ→ αBβ** **B→OB** **G2: A →0A|1A|00B** **B→OB|1B|Є** **G3:S →S10|S11|C|Є**	1 3	CO2	C1 C2 C3
	(c) What language is generated by the following grammar: 1. S→0S1| 01 2. S→aAb| bAb A→aA|bA|Є 3. S→E0E0E0 E→0E|1E|Є	3	CO2	C2
2	(a) Consider the following context-free grammar: S → S S + | S S * | a Show how the string **aaa+a*+aa**** can be generated by this grammar. Construct a parse tree for the string.	3	CO2	C4
	(b) Demonstrate briefly different phases of compiler by considering following example. Point out outputs of each phases with figure. **Percentage = Number * Percent /100;**	3	CO1	C2
	(d) Write the differences between compiler and interpreter.	1		C1
	Describe whether the following grammar is ambiguous or not by considering the string: "8 – 4 – 2+1". If the grammar is ambiguous how can you make it unambiguous? S → S + S | S – S |0|1|2|3|4|5|6|7|8|9	3	CO2	C4
3	(a) Construct a DFA and Є-NFA for the following regular expression with step by step clarifications: **(a|b)* abb (a|b)***	4	CO2	C3
	(b) What is *symbol table*? Explain the role of symbol table in each phase.	3	CO1	C2
	(c) Construct production rules of a grammar that describes the following language: *"List of digits separated by plus or minus signs"*	3	CO2	C4
4	(a) Define *token, lexeme* and *pattern.* Give the tokens, lexemes and pattern for the following statement: **a = b * c – d;**	1 2	CO2	C1 C3
	(b) Convert following NFA to DFA and test generated DFA using proper string: 	4	CO2	C4
	(c) Describe the role of *analysis* and *synthesis* part of a compiler exploring all stages of them.	3	CO1	C2

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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

Mid-Term Examination, Spring 2019

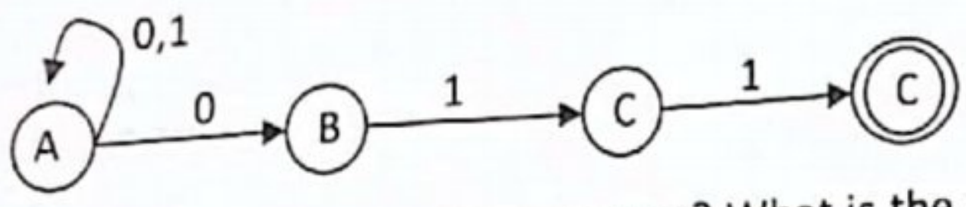
Course Code: **CSE-4801** Course Title: **Compiler**

Full Marks: **30** Time: **1 Hour 30 minutes**

*[Answer any **three** of the following questions (Figures at right margin illustrate marks)]*

No.	Questions	Marks
1. a)	Illustrate the block diagram to define a tool that used to convert a source program into target program.	2
b)	Write the similarities of compiler with Natural Language Processing (NLP).	1
c)	Draw the diagram of compiler's phases with output annotation of each phase to translate the following statement of C programming language. `net_salary:=gross_salary-basic_salary*tax_rate;`	5
d)	Show the front-end and back-end of compiler with appropriate figure.	2
2. a)	What do you mean by ambiguity? Justify whether the following grammar is ambiguous or not for input string **"9-5+2-1".** **string → string + string |string − string | 0|1|2|3|4|5|6|7|8|9|0**	3
b)	Elucidate the closer (*) rules of regular expression with example.	2

c)	Show the roles of lexical analyzer with block diagram. Identify the tokens, lexemes and patterns of the following C programming statements in a table:
	`long factorial(int n){`
	`  long fact=1;`
	`  for(int i=1;i<=n;i++)`
	`  fact*=i;`
	`  return fact;`
	`}`
		4

d)	What is the basic difference between DFA and NFA?	1
3. a)	What is language processing system? Illustrate the analysis of source program in a compiler.	3
b)	What is DFA? Convert the following NFA to DFA.  	4
c)	What are the main components of a grammar? What is the main factor to identify the grammar type? Classify the following grammar as regular, right-linear, left-linear, context free or context sensitive. Some grammar may fit more than one category. **G1:** **S → aM** **M → NP** **Ma → b** **Pc → c** **G2:** $S \rightarrow aX \mid \varepsilon$ $X \rightarrow aY$ $Y \rightarrow b$ **G3:** $A \rightarrow aA \mid \varepsilon$ $B \rightarrow Ab \mid b$ $C \rightarrow c$	3
d)	What is the basic difference between DFA and NFA?	1

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4. a)	Design a regular expression to define the number of C programming language.	2
b)	i). Define regular expression for all string of 0's and 1's that contain 01	2
	ii). Describe the language denoted by the regular expression $(a\|b)^*abb$	
c)	Write the regular expression to define the identifier of C programming language and construct the $\epsilon - NFA$ and DFA by using that regular expression.	3
d)	Consider the following CFG: $E \rightarrow E$ or $T\|T$ $T \rightarrow T$ and $F\|F$ $F \rightarrow$ not F/(E)/true/false i). Show that: "`not (true or false) and true`" can be generated by the grammar. ii). Construct the syntax tree and parse tree for the input sentence "`not (true or false) and true`".	3

END

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International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Mid Term Examination, Spring 2020

Course Code: CSE-3527 **Course Title:** Compiler

Total Marks: 21 Time: 180 minutes

Answer all 3 from the following Questions. Figures in the right-hand margin indicate full marks

1	```
	1. #include <iostream>
	2. using namespace std;
	3.
	4. int main() {
	5. float english, math, physics, total;
	6. int CT;
	7. cin >> english >> math >> physics >> CT;
	8. total = english + math + physics + CT * 2 / 3;
	9.
	10. if(total > 40){
	11. cout << "Pass!" << "\n";
	12. }else{
	13. cout << "Fail!"<< "\n";
	14. }
	15.
	16. return 0;
	17. }
	``
	a. Perform lexical analysis for the line number 5,6,7,8 in the code.
	b. Perform syntax, sematic and intermediate code generation for line number 8.
	c. Perform code optimization and code generation phases for 1(b).
		2.5
		2.5
		2
			C3	CO2

2	a. Make production rules for the following statement: "In a string of symbols, a, b and c, power of the b must be equal to the summation of the power of a and c" b. Using following CFG, S → S ∪ S | SS | S* | (S) | 0 | 1 | Ø | e give a derivation and the corresponding parse tree for the string **((1∪0(1*1))(1*1))** c. Describe the variation of grammar. Determine and describe the type of grammar for the string in the question 2(b) and justify your answer by proper examples.	3 2 2	C2 C3 C3	C02
3	a. Find out Q, q, F, δ and state diagram for the following language where Σ = {0,1}. And test your machine using a proper string. {w : 110 is not a substring of w} b. Give regular expressions describing the following languages. In all cases, the alphabet is {0, 1}. {w : w contains at least two 1s and at most one 0}	2.5 1.5 3	C3 C3 C3	C02

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International Islamic University Chittagong

Department of Computer Science & Engineering
B.Sc. in CSE, Mid Term Examination, Spring 2022
Course Code: CSE-3527 **Course Title:** Compiler
Total Marks: 21 Time: 180 minutes

Answer all 3 from the following Questions. Figures in the right-hand margin indicate full marks

1

```
1.  #include <iostream>
2.  using namespace std;
3.
4.  int main() {
5.      int n, b;
6.      float a, m;
7.      cin >> a >> m >> n >> b;
8.      n=m+n*/10*b/2+a;
9.
10.     cout << n << endl;
11. }
```

a. Perform lexical analysis for the line number 5,6,7,8 in the code. 2.5

b. Perform syntax, sematic and intermediate code generation for line number 8. 2.5

c. Perform code optimization and code generation phases for 1(b). 2

2 a. Make production rules for the following statement:
"In a string of symbols, a and b, power of b is at least three and both of the symbols can sit in any combination" 3

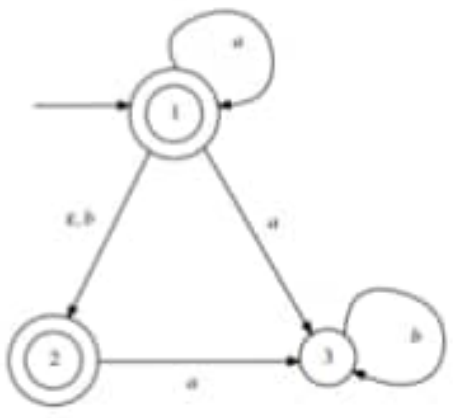
b. Using following CFG, S → S ∪ S | SS | S* | (S) | 0 | 1 | Ø | e 2
give a derivation and the corresponding parse tree for the string
((0*(1∪10)*0)*11)

c. Describe the variation of grammar. Determine and describe the type of grammar for the string in the question 2(b) and justify your answer by proper examples. 2

3 a. Find out Q, q, F, δ and state diagram for the following language where Σ = {0,1}. And test your machine using a proper string.
{w : w contains at least five 1s} 2.5

b. For each of the following regular expressions, give two strings that are members and two strings that are not members of the language described by the expression. The alphabet is Σ = {a, b}. 1.5
a(ba)*b*a(a∪b)* 3

c. Convert the following NFA to an equivalent DFA.



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International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE, Midterm Exam, 5th Semester, Spring 2022

Course Code: CSE 3527 **Course Title: Compiler**

Total marks: 30 Time: 1 hours 30 minutes

[Answer all the questions; in some questions, there are options;
Figures in the right-hand margin indicate full marks.]

1. a. Specify each phaseof a compiler explicitly with corresponding input and output while processing the following statement: 5

Expression := Term * factor + 128

b. Consider the grammar: 5

$$
\begin{aligned}
stmt &\rightarrow \textit{if-stmt} \mid \textbf{other} \\
\textit{if-stmt} &\rightarrow \textbf{if (}\ \textit{exp}\ \textbf{)}\ \textit{stmt} \\
&\mid \textbf{if (}\ \textit{exp}\ \textbf{)}\ \textit{stmt}\ \textbf{else}\ \textit{stmt} \\
\textit{exp} &\rightarrow 0 \mid 1
\end{aligned}
$$

i. Now, show that the grammar is ambiguous by constructing two parse trees for the following string:
if (0) if (1) other else other

ii. Rewrite the grammar so that it becomes unambiguous.

2. a. Consider the following grammar: 3

E→ E or T | T
T →T and F | F
F→ not F | (E) | true | false

Now, construct a parse tree for the following sequence:

not (true or false)

b. Suppose you are considering DFA from regular expression. Now, compute the *followpositions* for the following regular expression: 7
(a|b)*abb#ab

OR

Convert the regular expression R=(1|0)*1 to its equivalent NFA using Thompson construction.

3. a. i. What are the advantages of separating the analysis phase of compiling into lexical analysis and parsing? 5

ii. Why context free grammar is preferred over regular expressions in parsing?

b. Convert the regular expression R=(a|b)*abb(a|b) directly to its equivalent DFA. 5

OR

Prove that 0*(10*)* is equivalent to (1*0)*1*